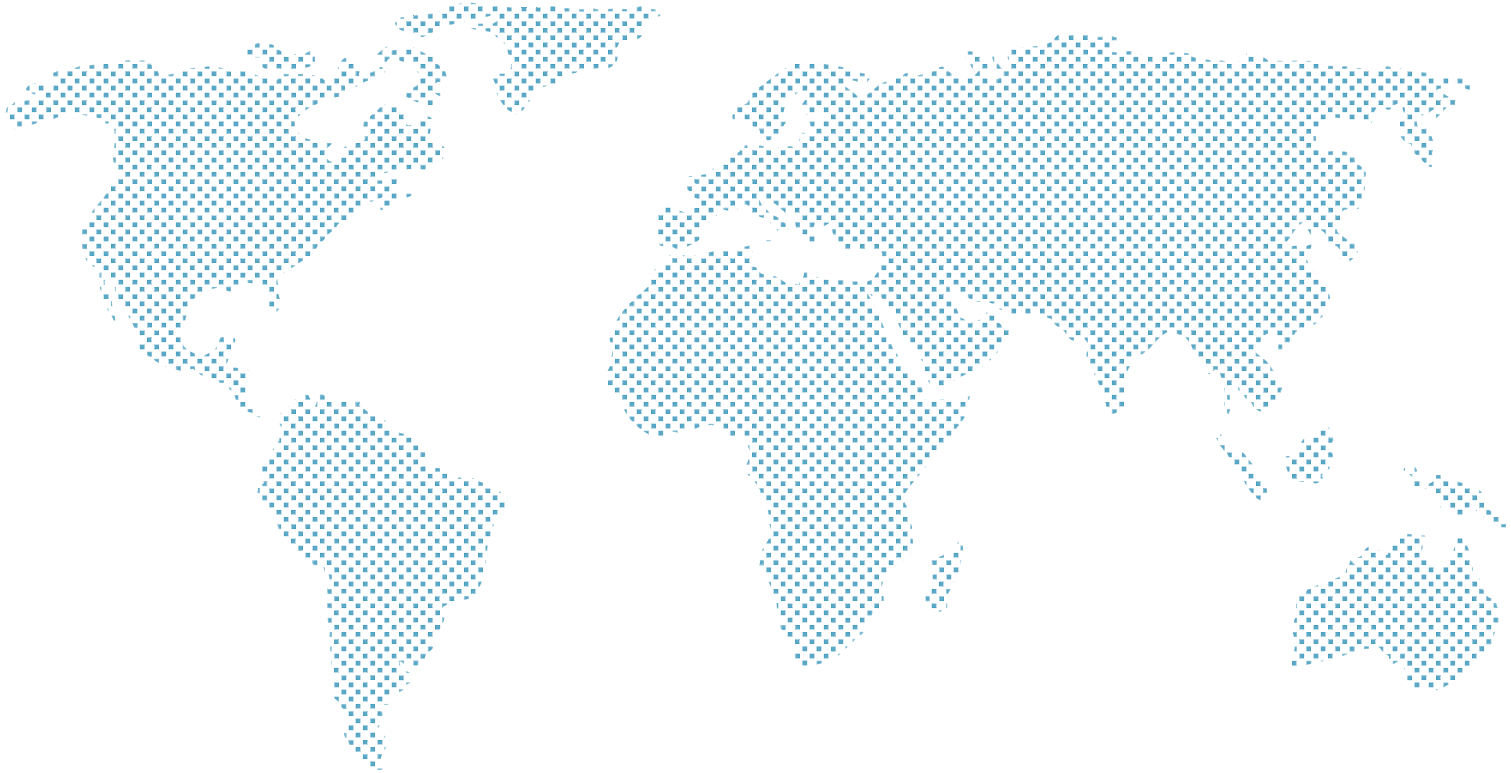




Document Purpose:

Discussion document to describe how text based AI Large Language Models (LLM) such as ChatGPT work and outline potential benefits and costs of getting started with self-hosted open source and third-party AI.

Chris Tyler – September 2025

Version	Issued By	Date	Comments
1.0	Chris Tyler	22 nd September 2025	Document Issued.
1.1	Chris Tyler	2 nd January 2026	Updates for 2026.

1. Overview

The aim is to discuss at 'high level' how AI Large Language Models such as [ChatGPT](#) work (see [Section 2](#)) and outline potential benefits and low-cost options for getting started with self-hosted and third-party AI in 2026.

AI Large Language Models (LLMs) can be viewed as 'Information Processing Toolkits'. These 'tools' can rapidly find, analyse, summarise, translate and generate information, helping users work with complex data, answer questions and create good quality content efficiently. This makes otherwise unstructured information such as documents stored in network folders more accessible and actionable (see [Section 3](#) for examples).

[ChatGPT](#), [Google Gemini](#) (and others) subscriptions to their latest models start from around £20 per month per user, but also offer 'free for personal use' versions, which are probably already being used by individuals within most organisations. For anyone else, these free versions are a good way to get an insight into current AI LLM capabilities and limitations.

[Microsoft 365 Copilot](#) adds AI capabilities to Word, Excel, Outlook and Teams, enabling users to analyse information, generate content, interpret data and convert unstructured material into usable outputs. It works [securely](#) with content in OneDrive, SharePoint and Teams, supporting summarisation, analysis, drafting and workflow automation. Organisations can also create custom Agents to automate multi-step processes and routine tasks. [Microsoft 365 Business](#) packages with Copilot start from around £20 per month per user.

Services such as [ChatLLM](#) and [OpenRouter](#) from around £10 per month, provide lost-cost access to all the latest models including ChatGPT, Google Gemini, Claude, Grok and many others. This price includes APIs which allow these models to be securely integrated within self-hosted and in-house applications. Assurances from individual providers over data privacy vary (e.g. [ChatLLM](#) and [OpenRouter](#)) and need to be understood.

Free to use 'open source' LLMs derived from [ChatGPT](#), [Google Gemini](#), [Microsoft](#) and many others, can be downloaded from reputable sites such as [Ollama Library](#) and [Huggingface](#) (see [Section 6](#)) and self-hosted.

Running AI LLMs locally does not need any internet or external 3rd party access. Smaller models such as [Llama 3.2](#) from Meta, are performant on most reasonably modern Desktop PCs, Laptops and Servers. Self-hosting offers cost effective, [private](#) access to AI Large Language Models (LLMs), as discussed further in [Section 4](#).

2. What are AI Large Language Models (LLMs) and how do they work?

Large Language Models (LLM) do not store facts, they store mathematical probabilities. [ChatGPT](#), [Google Gemini](#), [Microsoft Copilot](#) et al, are essentially pattern-matching systems that generate plausible, human-like content, by being trained to predict the next most likely word in a sentence, similar to a highly sophisticated autocomplete. For a visual explanation, watch this 7 min video [AI Large Language Models Explained Briefly](#).

Large Language Models (LLMs) are trained on vast amounts of text from books and the internet, allowing them to learn complex grammar, context and factual patterns. However, LLMs do not have human intuitions.

To strengthen their position and despite the very high costs of developing and training their products, AI companies such as [Meta](#), [Google](#), [Open AI \(ChatGPT\)](#) and others, release free to use 'open source' versions of their LLMs, to drive widespread community adoption of their models and strengthen their market position, by encouraging developers to build on their platforms.

Thus, community versions of [ChatGPT](#), [Google Gemini](#) and [Microsoft Copilot](#) are free for personal use, without needing an account or logging in. These are very usable despite not being the latest models. Subscriptions to the latest and 'best in class' cloud hosted models, typically cost between £10 and £20 per month per user.

Although full size enterprise LLM models such as [ChatGPT](#) and [Google Gemini](#) are enormous, perfectly usable versions such as [Llama 3.2](#) from Meta, have been shrunk to as small as 2GB. This is accomplished through sophisticated mathematical techniques that act like data compression, removing redundant information and [simplifying the model's structure without significant loss of performance](#).

3. Tasks for self-hosted AI large Language Models?

The videos linked below, demonstrate the general setup and capabilities for some specific examples.

1. **Process and Summarise Documents (e.g. Contracts, Policies, Procedures, Reports, Technical Docs)**
 - Load company or 3rd party documents and ask questions in natural language.
 - **Demo:** [Feed Documents to a Local Large Language Model](#) (first 2mins are different topic)
 - Video demonstrates explains the two options for feeding documents to LLMs – via the Context Window and Retrieval Augmented Generation (RAG).
 - Web sites can also be scanned for information.
2. **Drafting Documents on any topic.**
 - Example: Try “*We are a small business which holds some customer data. Draft a 20 point checklist for an upcoming GDPR assessment.*” on [ChatGPT](#), [Google Gemini](#), or [Ollama](#) running local LLMs on a Desktop or Laptop PC – including using voice rather than the keyboard.
3. **Brainstorm Ideas and Strategies**
 - Use any LLM as a creative partner to explore new ideas or strategies.
4. **Create Meeting Agendas and Summaries**
 - Provide a few bullet points and have the LLM flesh out a full meeting agenda, or paste raw notes, audio transcripts and ask for a clean summary.
5. **Microsoft 365 Copilot – Excel, Outlook, PowerPoint, Teams, Word**
 - **Overview:** [Microsoft Copilot](#)
 - **Excel:** [Copilot Excel Tutorial](#)
 - **Word:** [Copilot in Word | Transform a document](#)
 - **Outlook and Teams:** [Outlook and Teams: Enhance Communications](#)

4. Self Hosting – Benefits, Hardware and Security.

Self-hosted LLMs will not outperform premium AI services from [Meta](#), [Google](#), [OpenAI \(ChatGPT\)](#), [Microsoft](#) and others for quality. However, provided each self-hosted solution is ‘good enough’ for its task(s), being able to state that no client or confidential documents/information passes outside of your organisation is valuable.

Costs need not be high for small organisations or personal use.

1. For individual users, reasonably modern Desktop or Laptop computers with at least 8GB of system RAM, will run [Llama 3.2](#) from Meta and some other small LLMs listed in [Section 5](#) below. [Ollama](#), [LMStudio](#) and [AnythingLLM](#) (for building AI Agents) are easy to use and good for getting started.
2. If performance is an issue on a particular PC, a £200 Nvidia Graphics Card (GPU) with at least 6GB will work for models whose ‘physical size’ is smaller than the GPU memory size.
3. For teams and departments, LLMs can be securely self-hosted on Servers using Open Source software e.g., [Ollama](#), [Open-WebUI](#), [Searxng](#) and [Stable Diffusion](#) (for multi-media), all running under [Docker](#).
4. For Servers, graphics cards will be required but again, consumer grade Nvidia Graphics cards such as the RTX 3060 12GB and above, will be sufficient provided there is sufficient total GPU memory to comfortably hold all the LLMs that are being used in any given moment.

5. Selection of free to download and use Large Language Models.

The models listed below can all be downloaded from reputable community libraries such as [Ollama Library](#) and [Hugging Face](#).

Model Name	Size (Parameters)	Physical Size (GB)	Producer	Generates Images/Video	Notes
Qwen 1.5 7B	7B	4.6	Alibaba Cloud	No	Strong multilingual model, excellent for global business communications and content localization.
Qwen 1.5 14B	14B	8.3	Alibaba Cloud	No	Larger version with a 32k context window, suitable for analysing long documents and reports.
AnimateDiff v3	1.1B	5.0	ByteDance / Community	Yes - Video	A motion module used with image models (like Stable Diffusion 1.5) to create text-to-video animations. Very VRAM-efficient.
Gemma 2 9B	9B	5.4	Google	No	A powerful and versatile model from Google, suitable for a wide range of business tasks.
Gemma 7B	7B	4.8	Google	No	Lightweight and efficient; great for sentiment analysis, text summarization, and customer support chatbots.
Llama 3 8B	8B	4.7	Meta	No	Excellent general-purpose model with top-tier performance in reasoning, coding, and instruction following.
Code Llama 13B	13B	7.3	Meta	No	A specialized coding assistant that can significantly speed up software development and debugging cycles.
Phi-3 Medium	14B	7.9	Microsoft	No	A powerful and versatile model that offers a great balance of performance and efficiency for general tasks.
Phi-3 Mini	3.8B	2.3	Microsoft	No	A very lightweight yet capable model designed to run on-device, ideal for mobile and edge computing apps.
Mistral 7B	7B	4.1	Mistral AI	No	High-performing and efficient, known for strong reasoning and language understanding.
Yi 1.5 9B	9B	5.2	01.AI	No	A high-performing open-source model with strong bilingual (English/Chinese) capabilities.
Playground v2.5	2.6B	6.9	Playground AI	Yes - Image	Creates aesthetically pleasing images. Perfect for marketing, social media, and product mock-ups.
Stable Diffusion 1.5	0.9B	4.3	Stability AI	Yes - Image	A fast and lightweight model for rapid image generation. Great for iterative design and concept art.
Stable Diffusion XL	2.6B	6.9	Stability AI	Yes - Image	The flagship image model. Produces high-resolution, detailed images in a wide variety of styles.
Stable Video Diffusion	2.2B	5.2	Stability AI	Yes - Video	Generates short video clips from a starting image. Useful for creating dynamic ads, GIFs, and animated content.